

mean medium mode

```
# mean
marks = [45,50,55,60,90]
mean = sum(marks) / len(marks)
print("Mean:",mean)

Mean: 60.0

#median
import statistics
marks = [45,50,55,60,90]
median = statistics.median(marks)
print("Median",median)

Median 55

# mode
import statistics
data = [10,20,20,30,40]
mode = statistics.mode(data)
print("Mode:",mode)

Mode: 20

## variance code
import numpy as np
data = [10,12,14,16,18]
print(np.mean(data))
variance = np.var(data)
print("Variance:",variance)

14.0
Variance: 8.0

import pandas as pd
df=pd.DataFrame({
    'Marks': [45,50,55,60,90]
})
print("Mean:")
print(df["Marks"].mean())
print("\nMedian:")
print(df['Marks'].median())
print("\nMode:")
print(df['Marks'].mode())

Mean:
60.0
```

```
Median:
55.0

Mode:
0      45
1      50
2      55
3      60
4      90
Name: Marks, dtype: int64
```

Variance

```
import numpy as np
data = [10,12,14,16,18]
variance = np.var(data)
std=np.std(data)
print("Variance:",variance)
print("std:",std)

Variance: 8.0
std: 2.8284271247461903
```

IQR code

```
import numpy as np
data=[5,10,15,20,25,30,35,40]
Q1 = np.percentile(data,25)
Q3 = np.percentile(data,75)
IQR = Q3 - Q1
print("Q1:",Q1)
print("Q3:",Q3)
print("IQR:",IQR)

Q1: 13.75
Q3: 31.25
IQR: 17.5

#code
import numpy as np
data=[5,10,15,20,25,30,35,1000]
Q1 = np.percentile(data,25)
Q3 = np.percentile(data,75)
IQR = Q3 - Q1
lower = Q3 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
```

```

print(lower)
print(upper)
outliers = [x for x in data if x < lower or x > upper]
print("Outliers:",outliers)

```

```

-12.5
57.5
Outliers: [1000]

```

```

sample_space = {1,2,3,4,5,6}
event_A = {2,4,6}
print("Sample space(s):",sample_space)
print("Event A:",event_A)

```

probability p(a)

```

P_A = len(event_A) / len(sample_space)
print("\nProbability of Event A")
print("P(A) =", P_A)

```

```

Sample space(s): {1, 2, 3, 4, 5, 6}
Event A: {2, 4, 6}

```

```

Probability of Event A
P(A) = 0.5

```

```

P_Disease=0.01
P_positive_given_Disease=0.95
P_positive=0.05
P_disease_given_positive=(
    P_positive_given_Disease*P_Disease)/P_positive
print(P_disease_given_positive)

```

```

0.18999999999999997

```

conditional probability

```

event_A = {2,4,6}
event_B = {4,5,6}
intersection = event_A.intersection(event_B)
P_A_give_B = len(intersection) / len(event_B)
print("\nEvent B:",event_B)
print("A intersection b:",intersection)
print("P(A|B) =",P_A_give_B)

```

```

Event B: {4, 5, 6}
A intersection b: {4, 6}
P(A|B) = 0.6666666666666666

```

```

#Bayes' theorem
'''
example:
Disease prevalence=1%
test sensitivity=99%
positive test probability=5%
'''
P_Disease=0.01
P_positive_given_Disease=0.95
P_positive=0.05
P_disease_given_positive=(
    P_positive_given_Disease*P_Disease)/P_positive
print("Bayes'theorem example")
print("P(disease) =",P_Disease)
print("P(Positive|Disease) =",P_positive_given_Disease)
print("P(Positive)=",P_positive)
print("P(Disease|Positive)=",
    round(P_disease_given_positive,4)
)
print("Percentage=",
    round(P_disease_given_positive*100,2),
    "%")

print(P_disease_given_positive)

Bayes'theorem example
P(disease) = 0.01
P(Positive|Disease) = 0.95
P(Positive)= 0.05
P(Disease|Positive)= 0.19
Percentage= 19.0 %
0.18999999999999997

import pandas as pd
df= pd. DataFrame({
    'Hours' : [1,2,3,4,5],
    'Marks' : [20,40,50,70,90]
})
corr = df['Hours'].corr(df['Marks'])
print("Pearson Correlation:",corr)
# close to +1
#strong positive relation

Pearson Correlation: 0.9948497511671097

#spearman correlation
import pandas as pd
df= pd. DataFrame({
    'Rank_Study' : [1,2,3,4,5],
    'Rank_Marks' : [2,1,3,5,4]
})

```

```

}))
corr = df['Rank_Study'].corr(
    df['Rank_Marks'],
    method='spearman'
)
print("Spearman Correlation:",corr)

```

Spearman Correlation: 0.7999999999999999

```

#matrix correlation
import pandas as pd
df= pd. DataFrame({
    'Hours' : [1,2,3,4,5],
    'Marks' : [20,40,50,70,90],
    'Sleep' : [8,7,6,5,4]
})
print(df.corr())

```

	Hours	Marks	Sleep
Hours	1.00000	0.99485	-1.00000
Marks	0.99485	1.00000	-0.99485
Sleep	-1.00000	-0.99485	1.00000

```

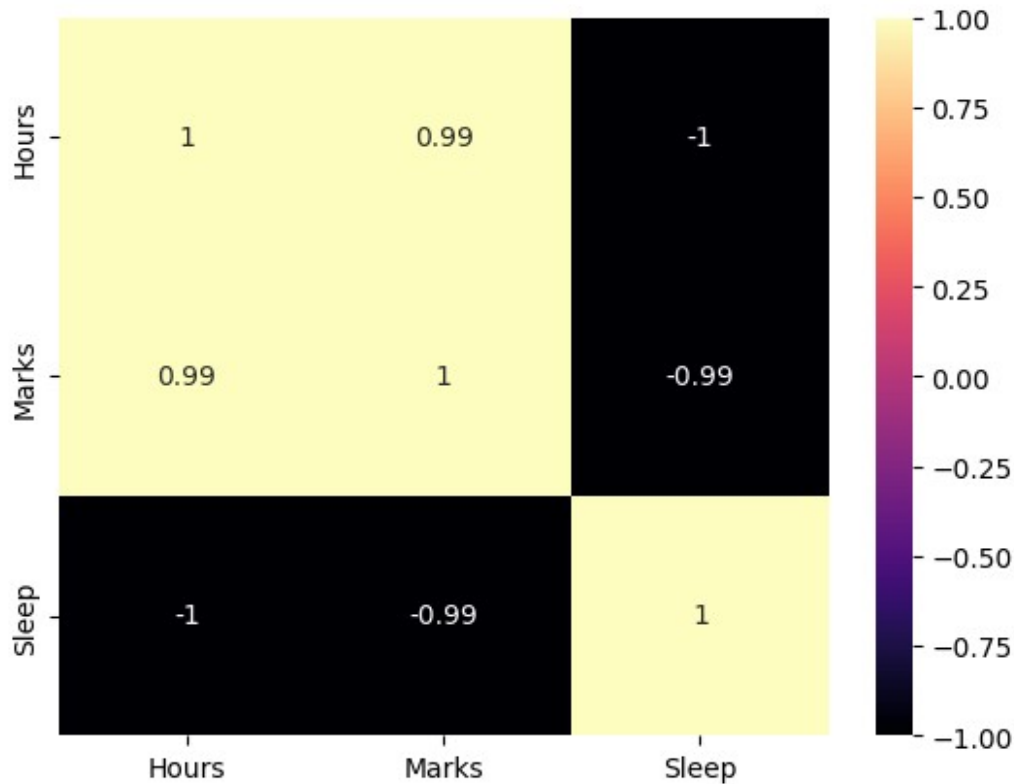
##heatmap
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df= pd. DataFrame({
    'Hours' : [1,2,3,4,5],
    'Marks' : [20,40,50,70,90],
    'Sleep' : [8,7,6,5,4]
})
sns.heatmap(df.corr(),
annot=True,
cmap='magma')
plt.show()

```

```

#common cmap values
'''
1 coolwarm
2 viridis
3 YLGnBu
4 blues
5 red
6 greens
7 magma
'''

```

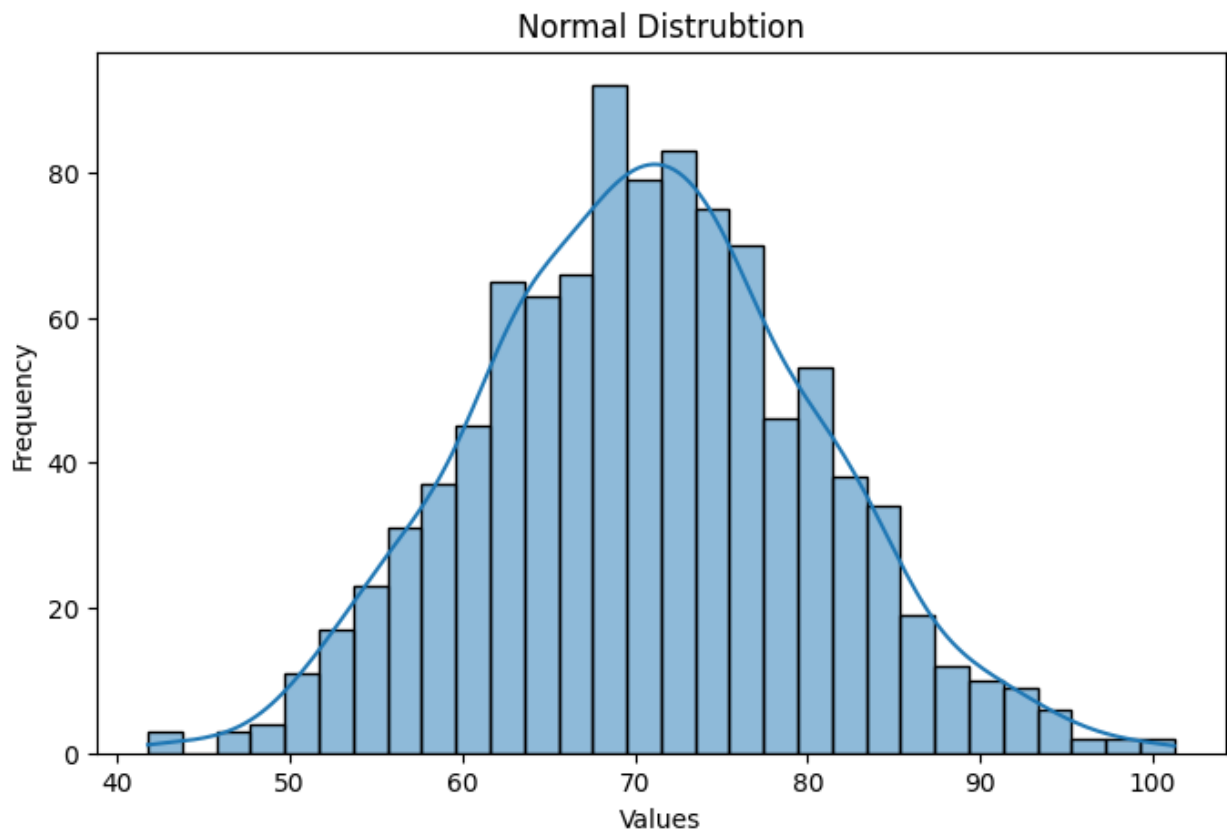


Normal distribution

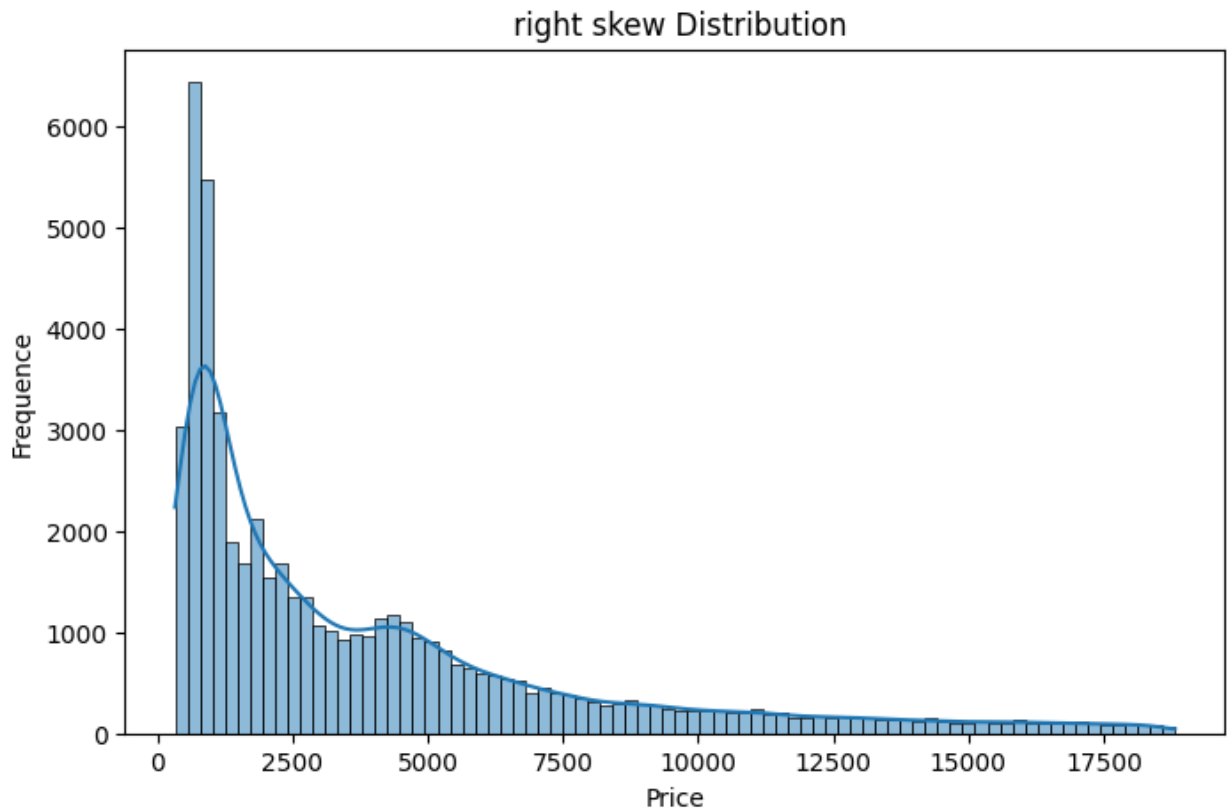
```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
#generate normally distributed data
data = np.random.normal(
    loc=70,      #mean
    scale=10,    #standard deviation
    size=1000    #number of values
)
#calculate mean and standard deviation
mean = np.mean(data)
std = np.std(data)
print("Mean:", round(mean, 2))
print("Standard Deviation:", round(std, 2))

plt.figure(figsize=(8, 5))
sns.histplot(data, bins=30, kde=True)
plt.title("Normal Distrubtion")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()
```

Mean: 70.65
Standard Deviation: 9.66

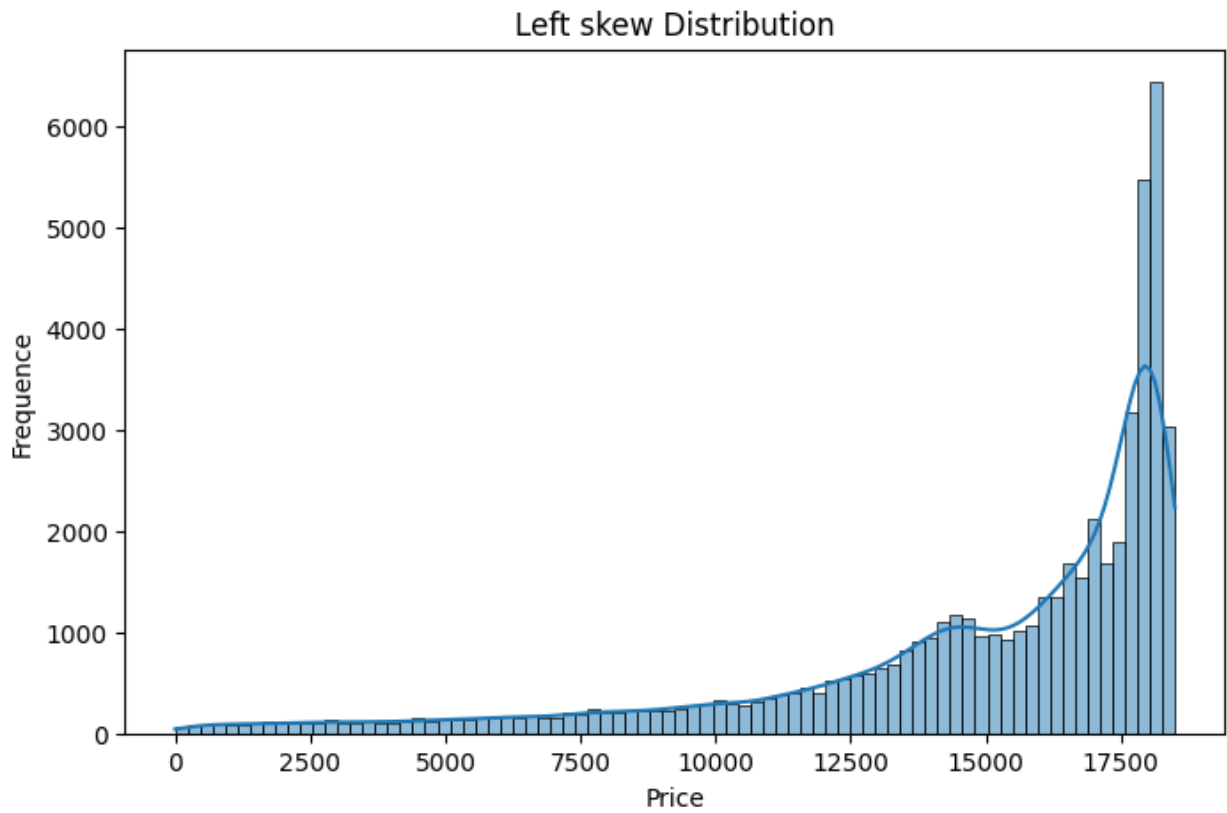


```
import seaborn as sns
import matplotlib.pyplot as plt
#load Dataset
df=sns.load_dataset("diamonds")
#histogram + KDE
plt.figure(figsize=(8,5))
sns.histplot(df["price"],kde=True)
plt.title(" right skew Distribution")
plt.xlabel("Price")
plt.ylabel("Frequence")
plt.show()
```



```
import seaborn as sns
import matplotlib.pyplot as plt
#load Dataset
df=sns.load_dataset("diamonds")

#create left-skewed data
left_skew = df["price"].max()-df["price"]
#histogram + KDE
plt.figure(figsize=(8,5))
sns.histplot(left_skew,kde=True)
plt.title("Left skew Distribution")
plt.xlabel("Price")
plt.ylabel("Frequency")
plt.show()
print("Skewness:",left_skew.skew())
```

Skewness: -1.6183952833835282